

NIST CSF 2.0

LearnSphere EdTech Cybersecurity Gap Assessment — Target Maturity Profile

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Organization	LearnSphere EdTech	Framework	NIST CSF 2.0

Company / Entity	LearnSphere EdTech
Standard Reference	NIST Cybersecurity Framework (CSF) 2.0; NIST Quick-Start Guide: Creating and Using Organizational Profiles
Related Documents	Project 1 - LearnSphere EdTech ISMS (ISO/IEC 27001:2022 implementation documentation set: ISMS scope, asset register, risk register, Statement of
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Reviewed By	Information Security Manager
Approved By	Chief Information Security Officer (CISO)
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1. Target-Setting Methodology

Target maturity is set against the Current ratings established in Step 3. The standard rule is **Current + 1**, moving LearnSphere EdTech toward defined and consistently implemented outcomes without assuming immediate optimization.

Exception — PR.AA-05: the target is Level 4 rather than Level 3 because student records are highly sensitive and access governance is a critical risk area. The target therefore requires managed measurement and continuous improvement of least privilege, separation of duties, privileged access and access-review effectiveness.

Targets reflect the actual sector and operating profile: an AWS-hosted LMS processing student information and academic records, including information relating to minors, with material exposure around student-data confidentiality, access control, data lifecycle and privacy governance.

2. Function-Level Target Summary

Function	Current	Target	Gap	Priority
Govern	1.75	3.00	1.25	High
Identify	1.75	3.00	1.25	High
Protect	1.75	3.25	1.50	Critical
Detect	1.75	3.00	1.25	High
Respond	2.00	3.00	1.00	Medium-High
Recover	1.75	3.00	1.25	Medium

3. Govern (GV) — Target Maturity

Function target average: 3.00 | Priority: High

Subcategory	Current	Target	Gap	Target-State Rationale
GV.OC-03	1	3	2	Document and consistently implement privacy, children's-data, parental-consent, purpose-limitation and related requirements.
GV.RM-01	2	3	1	Make cybersecurity/privacy risk assessment a defined trigger for new student-data processing and product changes.
GV.RR-02	2	3	1	Formalize accountable ownership across security, product/engineering and relevant privacy responsibilities.
GV.PO-01	2	3	1	Translate policy into consistently implemented collection, consent, minimization, access and retention requirements.

4. Identify (ID) — Target Maturity

Function target average: 3.00 | Priority: High

Subcategory	Current	Target	Gap	Target-State Rationale
ID.AM-02	2	3	1	Maintain a governed inventory of software/services and link relevant services to the student-data processing they support.
ID.AM-07	1	3	2	Create an authoritative student-data inventory with owner, purpose, location, sensitivity, flow and retention metadata.
ID.RA-04	2	3	1	Establish a repeatable risk/privacy assessment trigger for changes to student-data processing and product functionality.
ID.RA-05	2	3	1	Make data-minimization risk an explicit, documented input to risk assessment for new or changed processing.

5. Protect (PR) — Target Maturity

Function target average: 3.25 | Priority: Critical

Subcategory	Current	Target	Gap	Target-State Rationale
PR.AA-05	2	4	2	Use managed, measured and continuously improved RBAC, least privilege, separation of duties, privileged-access certification and access-review metrics.
PR.AA-03	2	3	1	Define consistent authentication requirements appropriate to sensitive student-record and privileged access.
PR.DS-01	2	3	1	Document and consistently implement protection of student data at rest across production databases, backups and relevant repositories.
PR.DS-10	1	3	2	Establish enforceable retention schedules, secure deletion, exception handling and disposal verification.

6. Detect (DE) — Target Maturity

Function target average: 3.00 | Priority: High

Subcategory	Current	Target	Gap	Target-State Rationale
DE.CM-01	2	3	1	Define monitoring coverage requirements for critical AWS, LMS and student-data environments and maintain repeatable evidence.
DE.CM-06	2	3	1	Formalize monitoring expectations and evidence requirements for relevant external service providers.
DE.AE-02	2	3	1	Standardize alert investigation, evidence capture and operational metrics such as MTTD/MTTI.
DE.AE-04	1	3	2	Define repeatable correlation and impact-assessment use cases linking IAM, application activity and student-record access.

7. Respond (RS) — Target Maturity

Function target average: 3.00 | Priority: Medium-High

Subcategory	Current	Target	Gap	Target-State Rationale
RS.MA-01	2	3	1	Define incident criteria that consistently identify and prioritize events involving children's or student data.
RS.AN-03	2	3	1	Standardize evidence collection and analysis for suspected unauthorized student-record access.
RS.CO-02	2	3	1	Define communication roles, escalation paths and decision criteria for student-data incidents.
RS.MI-01	2	3	1	Establish documented playbooks for containment of compromised student records and related access scenarios.

8. Recover (RC) — Target Maturity

Function target average: 3.00 | Priority: Medium

Subcategory	Current	Target	Gap	Target-State Rationale
RC.RP-01	2	3	1	Conduct documented, repeatable recovery exercises for ransomware, database compromise and student-data loss.
RC.RP-03	2	3	1	Formalize recovery priorities, dependencies and RTO/RPO expectations for the LMS and critical student-data services.
RC.CO-02	2	3	1	Define repeatable recovery communications for student-service disruption and recovery milestones.
RC.IM-01	1	3	2	Create a formal lessons-learned process that assigns, tracks and verifies improvement actions.

10. Overall Target-State Conclusion

Overall target maturity: 3.04 / 4.00. Most outcomes are intentionally targeted at Level 3, providing a realistic progression from the current posture toward documented, assigned and consistently implemented practices.

The Protect Function has the highest target average at **3.25**. PR.AA-05 is the single Level 4 target because student-record access is a critical risk area and requires ongoing measurement and continuous improvement rather than one-time control implementation.

These Target ratings establish the baseline for Step 5 Gap Analysis, where **Gap = Target – Current** and the largest gaps are prioritized for remediation.

11. Approval

ROLE	NAME	SIGNATURE	DATE
Prepared By	Mr. Meejan Shaikh – GRC Analyst	_____	07-Sep-2026
Reviewed By	Information Security Manager	_____	07-Sep-2026
Approved By	CISO	_____	07-Sep-2026